# Irritant Contact Dermatitis 15

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**Synonyms:** ▪ Irritant reaction ▪ Irritant dermatitis ▪ Irritant dermatitis syndrome ▪ Irritant contact dermatitis syndrome ▪ Toxic contact dermatitis ▪ "Irritation" ▪ "Chemical burn"

## Key features

- Irritant contact dermatitis (ICD) is a localized, non-immunologically initiated, cutaneous inflammatory reaction. Its clinical characteristics are polymorphous and include erythema, scaling, edema, vesiculation, and erosions in acute cases, with erythema, lichenification, hyperkeratosis, and fissures in chronic cases
- ICD results from a direct cytotoxic effect due to single or repeated application of a chemical substance or physical insult to a cutaneous site
- ICD is a common disorder and represents an important spectrum of clinical presentations in both general and occupational dermatology, occurring when the normal epidermal barrier is disrupted and secondary inflammation develops
- Despite different pathogeneses, allergic contact dermatitis and ICD have similar clinical appearances, especially their chronic variants
- ICD is a multifactorial syndrome determined by the properties of the irritating substance as well as host and environmental factors; a dose–response relationship is related to exposure parameters such as concentration, pH, temperature, occlusion, repetition, and duration of contact
- Strong or absolute irritants such as strong acids and alkalis, as well as oxidants, produce erythema, edema, vesicles that may coalesce into bullae, oozing, and, in severe cases, necrosis and ulceration. The latter is vernacularly termed a "chemical burn"
- ICD is most often caused by cumulative exposure to weak irritants such as soap and water
- The clinical features of chronic ICD from repeated exposure to low-grade or marginal irritants (e.g. soaps, solvents, cleansers, weak acids and bases) include erythema, lichenification, excoriations, scaling, and fissures

## INTRODUCTION

There are two major forms of contact dermatitis: irritant and allergic. Irritant contact dermatitis (ICD) is a cutaneous inflammatory disorder resulting from activation of the innate immune system by the direct cytotoxic effect of a chemical or physical agent, whereas allergic contact dermatitis is a delayed-type hypersensitivity immune reaction mediated by hapten-specific T cells[1].

Contact dermatitis is a matter of public health importance[1], with total direct costs exceeding an estimated $1.5 billion in the US in 2013[2]. Occupational contact dermatitis represents a significant cost to society when accounting for medical costs, workers' compensation, decreased productivity, and lost time from work[3–6]. A National Health Interview Survey suggested that in 2010, 15 million Americans were affected by occupational dermatitis[6], a figure 100-fold higher than in prior reports by the US Bureau of Labor Statistics. Overall, the impact of occupational skin diseases on both society and the individual is formidable in light of both prevalence and economic factors.

## HISTORY

One of the earliest documented irritant skin reactions is in the writings of Celsus around 100 CE, who described skin ulceration resulting from corrosive metals[7]. The first accounts of occupational skin diseases focused on ill health among miners, and in 1556, Georg Agricola detailed the deep ulcers observed among metal workers. Eleven years later, Paracelsus discussed the etiology, pathogenesis, and therapy of skin changes caused by salt compounds. In 1700, Bernardino Ramazzini published a detailed treatise on the diseases of tradespeople in which he described skin disorders incurred by bath attendants, bakers, gilders, midwives, millers, and miners[7]. Physicians began to more widely recognize occupational irritant dermatitis during the Industrial Revolution with the development of new materials and chemicals, both natural and synthetic, for industrial and household use[8]. Germany and France were the first to enact laws compensating workers for industrial skin diseases.

## EPIDEMIOLOGY

ICD is the most common form of occupational skin disease, estimated to constitute between 70% and 80% of all occupational skin disorders (Table 15.1)[9]. Data from the North American Contact Dermatitis Group indicate that in groups of patients with contact dermatitis, ICD (as a primary diagnosis) accounts for 9%–10% of those screened for allergic contact dermatitis via patch testing, and when hands are affected, ICD predominates over all other causes of contact dermatitis[10,11].

The US Bureau of Labor Statistics data show that occupational skin diseases accounted for a consistent 30%–45% of all patients with occupational illnesses from the 1970s through the mid-1980s. However, occupational skin disease rates have since fallen dramatically, e.g. from an average rate of 16.2 events/10 000 full-time workers in 1972 to 2.8 events in 2013[12].

Although skin disease accounted for only 17% of all recorded nonfatal occupational illnesses in 2007[13,14], it still ranks second and is only surpassed by musculoskeletal disorders. The incidence of occupational contact dermatitis in several other countries is similar to that in the US, with a range of 50 to 70 cases/100 000 workers/year[15]. In Europe, the decreasing incidence of occupational contact dermatitis has been attributed to more recently enacted legislation that limits occupational exposure to chemical agents, though a causal relationship has not been confirmed[16,17]. In 1996, the United Kingdom created a voluntary reporting system that combined information from the EPIDERM study with the occupational physicians' reporting activity (OPRA). The annual incidence of occupational contact dermatitis reported by dermatologists was 0.9 per 10 000 workers, and for occupational physicians, it was 3.1 per 10 000 workers[1,18].

In 2010, the National Institutes of Occupational Safety and Health sponsored the National Health Interview Survey which utilized a sample cohort of 27 157 individuals (to represent 229 million US civilian adults), the majority of whom were employed or recently employed[6]. The overall prevalence of dermatitis was 10.2%, ranging from 7.3% in those who never worked to 9.8% in currently employed workers to 11.8% in those recently, but not currently, employed.

In addition to manufacturing, industries with the highest non-adjusted and adjusted rates of dermatitis include healthcare and social assistance; arts, entertainment, and recreation; and accommodations and food services. Rubber chemicals, soaps and cleansers, wet work, resins, acrylics, and nickel are some of the more common sources for irritant and allergic contact dermatitis. Additional high-risk occupations are discussed in Chapter 16[1,18].



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| EXPOSURES TO IRRITANTS BY OCCUPATION | |
|---|---|
| **Occupation** | **Irritants** |
| Agriculture workers | Artificial fertilizers, disinfectants, pesticides, cleaners, gasoline, diesel oil, plants/grains |
| Artists | Solvents, clay, plaster |
| Automobile and aircraft industry workers | Solvents, cutting oils, paints, hand cleansers |
| Bakers and confectioners | Flour, detergents |
| Bartenders | Wet work, detergents |
| Bookbinders | Solvents, glues |
| Butchers | Detergents, meat |
| Cabinet makers and carpenters | Glues, detergents, thinners, solvents, wood preservatives |
| Cleaners | Detergents, solvents, wet work |
| Coal miners | Dust (coal, stone), wet conditions |
| Construction workers | Cement |
| Cooks and caterers | Detergents, vegetable juices, wet work |
| Dentists and dental technicians | Detergents, hand cleansers, wet work |
| Dry cleaners | Solvents |
| Electricians | Soldering fluxes |
| Electroplaters | Acids, alkalis |
| Floor layers | Solvents |
| Florists and gardeners | Manure, artificial fertilizers, pesticides, wet work |
| Hairdressers | Permanent wave solutions, shampoos, bleaching agents, wet work |
| Homemakers | Detergents, cleansers, foods, wet work |
| Hospital workers | Detergents, disinfectants, foods, wet work |
| Jewelers | Detergents, solvents |
| Mechanics | Oils, greases, gasoline, diesel fuel, cleaners, solvents |
| Metal workers | Cutting oils, solvents, hand cleansers |
| Nurses | Disinfectants, detergents, wet work |
| Office workers | Paper products, solvents in cleaners, adhesives, and photocopiers/toner |
| Painters | Solvents, thinners, wallpaper adhesives, hand cleansers |
| Photography industry workers | Solvents, wet work |
| Plastics workers | Solvents, acids, styrene, oxidizing agents |
| Printers | Solvents |
| Rubber workers | Solvents, talc, zinc stearate |
| Shoemakers | Solvents |
| Tannery workers | Acids, alkalis, reducing and oxidizing agents, wet work |
| Textile workers | Fibers, bleaching agents, solvents |
| Veterinarians and slaughterhouse workers | Disinfectants, wet work, animal entrails and secretions |

**Table 15.1 Exposures to irritants by occupation.** *Adapted from Tovar R, Leikin JB. Irritants and corrosives. Emerg Med Clin North Am 2015;33:117–31.*

Clinical manifestations of ICD are highly variable and determined by the properties of the irritating substance as well as host and environmental factors. These include concentration, pH, mechanical pressure, temperature, humidity, and duration of contact. Low ambient humidity and cold are important factors in decreasing the water content of the stratum corneum and, consequently, increasing the permeability to irritants such as soaps, detergents, acids, bases, and solvents. Cold alone may also reduce the plasticity of the horny layer, with consequent cracking of the stratum corneum. However, in one study, the application of cold had a protective effect on the development of ICD if applied during the provocative exposure, in this case to sodium lauryl sulfate[19]. Occlusion, excessive humidity, and maceration increase the water content of the stratum corneum and compromise the stratum corneum's protective lipid bilayers, with consequent enhanced percutaneous absorption of water-soluble substances (Fig. 15.1)[20]. In addition, irritated skin may become more susceptible to superimposed allergic sensitization.

Important predisposing characteristics of the individual include age, sex, pre-existing skin disease, anatomic region exposed, and sebaceous activity. There are age-associated changes in the skin that can alter the skin's response to irritants. Both infants and the elderly are more often affected by ICD because of their less robust epidermal barrier, and they also develop more severe symptoms. While skin irritation may be seen more often on the upper extremities of women than men, this higher prevalence of ICD may be due to increased frequency of exposure rather than inherent sex differences. Genetic factors also play a role in the development of ICD, as shown in studies with monozygotic twins[21]. Patients with a history of atopic dermatitis have a 13.5 times greater risk of developing occupational dermatitis[22], and a reduction in epidermal filaggrin can reduce the inflammatory threshold for irritants[23]. Lastly, the most commonly affected sites are exposed areas such as the hands and the face, with hand involvement seen in ~80% of patients and facial involvement in 10%[15]. Excessive exposure to water, soaps and detergents, common causes of ICD, play a critical role given that wet work (immersion in water for >2 hours, occlusive protective gear, and/or hand washing >20 times/day) represents one of the most important risk factors for developing irritant dermatitis[24].

## PATHOGENESIS

ICD is due to direct contact of a chemical or physical agent with the skin, which triggers activation of the innate immune response without the need for prior sensitization. Although the cellular mechanisms of ICD remain elusive, increasing evidence suggests that activated keratinocytes act as signal transducers in the control of host homeostatic responses to exogenous stimuli and they serve as key immunoregulators. An irritant chemical's ability to induce cutaneous damage may serve to trigger pattern recognition receptors (PRR), including Toll-like receptors, providing the necessary impetus to activate the innate immune system[25]. While other mediators such as prostaglandins, leukotrienes, and neuropeptides may possibly play a role, cytokines carry the most interest in ICD as they are the central mediators in T-cell recruitment and inflammation.

Several pathomechanisms have been associated with ICD, including denaturation of epidermal keratins, disruption of the permeability barrier (see Ch. 124), damage to cell membranes, and direct cytotoxic effects, with different mechanisms at work with different irritants

**Fig. 15.1 Bilateral irritant contact dermatitis of the feet and ankles due to chronic occlusive footwear.**



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| IRRITANTS AND MECHANISMS OF TOXICITY | |
|---|---|
| **Irritant** | **Mechanisms of toxicity** |
| Detergents | Barrier disruption, protein denaturation, membrane toxicity |
| Acids | Protein denaturation, cytotoxicity |
| Alcohols | Protein denaturation |
| Alkalis | Barrier lipid denaturation, cytotoxicity through cellular swelling |
| Oils | Disorganization of barrier lipids |
| Organic solvents | Solubilization of membrane lipids, membrane toxicity |
| Oxidants | Cytotoxicity |
| Reducing agents | Keratolysis |
| Water | If barrier is disrupted, cytotoxicity through swelling of viable epidermal cells |

**Table 15.2 Irritants and mechanisms of toxicity.**

(Table 15.2). The mechanisms involved in the acute and chronic phases of ICD are fundamentally different. Acute reactions involve direct cytotoxic damage to keratinocytes, while repeated exposures to solvents and surfactants cause slower damage to cell membranes by removing surface lipids and water-retaining substances.

The pathogenic pathway in the *acute phase* of ICD, common to many chemically unrelated irritants, begins with penetration of the irritant through the permeability barrier (stratum corneum), mild damage or stress to keratinocytes, and the release of mediators of inflammation with resultant T cell activation[20]. In this manner, once activation is initiated via epidermal cells, continuous T cell activation independent of the exogenous agent may be maintained and barrier dysfunction perpetuated[26]. Tumor necrosis factor (TNF) and interleukin (IL)-1α are the major mediators, and they are capable of inducing production of other cytokines, chemokines, and adhesion molecules, leading to leukocyte recruitment to the site. Specifically, TNF, IL-6, and IL-1β upregulate expression of intercellular adhesion molecule-1 (ICAM-1)[27]. This is a predominant feature of ICD. In addition, IL-1 receptor antagonist (IL-1RA) and IL-8 increase substantially after exposure to the common irritant sodium lauryl sulfate.

In the *chronic phase* of ICD, the stratum corneum barrier is disrupted. Damage to the stratum corneum lipids (which mediate barrier function) is associated with loss of cohesion of corneocytes, desquamation, and an increase in transepidermal water loss (TEWL). TEWL is one of the triggering stimuli that promote lipid synthesis, keratinocyte proliferation, and transient hyperkeratosis during the restoration of the cutaneous barrier. However, damage with a solvent can disrupt this protective mechanism by occlusion and blockage of water evaporation, thus halting lipid synthesis and barrier recovery. After chronic exposure, the result is increased epidermal turnover manifested clinically by the chronic eczematoid irritant reaction[28].

## CLINICAL FEATURES

Several different types of ICD have been described[29] (see below). Consequences of the myriad forms of ICD range from postinflammatory pigmentary changes to poorly healing ulcers (Table 15.3).

| CLINICAL FEATURES SUGGESTING AN IRRITANT OR TOXIC ETIOLOGY | | |
|---|---|---|
| **Clinical feature** | **Possible irritant or toxin** | |
| Ulcerations | Inorganic acids (chromic, hydrofluoric, nitric, hydrochloric, phosphoric, sulfuric)<br>Strong alkalis (calcium oxide, sodium hydroxide, potassium hydroxide, ammonium hydroxide, calcium hydroxide, sodium metasilicate, sodium silicate, potassium cyanide, trisodium phosphate, sodium carbonate, potassium carbonate)<br>Salts (arsenic trioxide, dichromates)<br>Solvents (acrylonitrile, carbon bisulfide)<br>Gases (ethylene oxide, acrylonitrile) | |
| Folliculitis or acneiform lesions | Oils and greases<br>Tar and asphalt<br>Chlorinated naphthalenes and polyhalogenated biphenyls<br>Arsenic trioxide<br>Glass fibers | |
| Miliaria | Occlusive clothing and overheating<br>Adhesive tape<br>Ultraviolet and infrared radiation<br>Aluminum chloride | |
| Pigmentary changes | | |
| • Hyperpigmentation | Any irritant or allergen, especially phototoxic agents such as psoralens, tar, asphalt, psoralen-containing plants<br>Metals, such as inorganic arsenic (systemically), silver, gold, bismuth, mercury<br>Radiation: ultraviolet, infrared, microwave, ionizing | |
| • Hypopigmentation | *p*-Tert-amylphenol | *p*-Cresol |
| | *p*-Tert-butylphenol | 3-Hydroxyanisole |
| | Hydroquinone (HQ) | Butylated hydroxyanisole |
| | Monobenzyl ether of HQ | 1-Tert-butyl-3, 4-catechol |
| | Monomethyl ether of HQ | 1-Isopropyl-3, 4-catechol |
| | *p*-Tert-catechol | 4-Hydroxypropriophenone |
| Alopecia | Borax<br>Chloroprene dimers | |
| Urticaria | Animals (arthropods, caterpillars, corals, jellyfish, moths, sea anemones)<br>*Myroxylon pereirae* (balsam of Peru)<br>Cosmetics (e.g. sorbic acid)<br>Drugs (alcohol [ethanol], benzocaine, camphor, capsaicin, chloroform, iodine, methyl salicylate, resorcinol, tar extracts)<br>Foods (cayenne pepper, fish, mustard, thyme)<br>Fragrances and flavoring agents (cinnamon oil, cinnamic acid and aldehyde, benzaldehyde)<br>Metals (cobalt)<br>Plants (seaweed, stinging nettles), woods<br>Preservatives (benzoic acid)<br>Textiles (e.g. blue dyes) | |

**Table 15.3 Clinical features suggesting an irritant or toxic etiology.**

### Acute Irritant Contact Dermatitis

*Acute* ICD, commonly seen with occupational accidents, develops when the skin is exposed to a potent irritant. The latter is capable of profound reactions on the skin surface: oxidation, reduction, desiccant, vesicant, or ion disruption[30]. The acute reaction reaches its peak quickly, usually within minutes to hours after exposure, and then starts to heal. This is termed the decrescendo phenomenon. Symptoms of acute ICD include burning, stinging, and soreness of the directly affected sites. Physical signs include erythema, edema, bullae, and possibly necrosis. These lesions are restricted to the area where the irritant or toxicant damaged the tissue, with sharply demarcated borders and asymmetry pointing to an exogenous cause. If there is no dermal injury, healing should be complete[31].

The potent irritants that most frequently lead to ICD are acids and alkalis, resulting in chemical burns.

### Acute Delayed Irritant Contact Dermatitis

*Acute delayed* ICD is a retarded inflammatory response characteristic of certain irritants, such as anthralin (dithranol), benzalkonium chloride (preservative/disinfectant), and ethylene oxide. Adverse reactions to these chemicals are considered idiosyncratic, except when they are applied to previously injured skin, e.g. sites of xerosis or atopic dermatitis[31]. Clinically visible inflammation is not seen until 8 to 24 hours (or more) after exposure, and thus may mimic allergic contact dermatitis; however, the associated symptom is more frequently burning rather than pruritus. Sensitivity to touch and water is elicited. This form of ICD is commonly seen during diagnostic patch testing.



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## Irritant Reaction Irritant Contact Dermatitis

*Irritant reaction* ICD is a type of subclinical irritant dermatitis in individuals exposed to wet chemical environments, such as hairdressers, caterers, metal workers, or those with frequent exposures to soap and water. It is characterized by one or more of the following signs: scaling, redness, vesicles, pustules, and erosions, often beginning under occlusive jewelry (e.g. rings) and then spreading onto the fingers and then the hands and the forearms. It may simulate dyshidrotic dermatitis and ultimately result in cumulative ICD if exposure is prolonged; however, ICD tends to resolve if exposure is discontinued.

## Cumulative Irritant Contact Dermatitis

*Cumulative* ICD is a consequence of multiple sub-threshold skin insults, without sufficient time between them for complete restoration of skin barrier function (see Fig. 16.4). It may be due to a variety of stimuli or frequent repetition of one factor, e.g. exposure to water both at the workplace and at home. Clinical symptoms develop only after the cumulative damage exceeds an individually determined manifestation or elicitation threshold, which may decrease with progression of the disease. Weak irritants do not lead to clinical ICD if they are encountered far enough apart to allow for restoration of skin barrier function. However, if the same irritant exposures follow each other closely in time, or when the manifestation threshold is reduced (e.g. in a patient with active atopic dermatitis), cumulative ICD can develop. The properties of the irritating substance, e.g. pH, solubility, detergent action, physical state, are also important. In contrast to acute ICD, the lesions of chronic ICD are less sharply demarcated. Pruritus and pain due to fissures of hyperkeratotic skin are symptoms of chronic ICD. Signs may include xerosis, erythema, and vesicles, but lichenification and hyperkeratosis predominate.

## Asteatotic Dermatitis

*Asteatotic dermatitis*, also referred to as asteatotic eczema, eczema craquelé, or exsiccation eczematid ICD, is a special variant seen primarily during dry winter months. Elderly individuals who frequently bathe without remoisturizing are at particular risk of developing asteatotic dermatitis. Intense pruritus is common, with the skin appearing dry with ichthyosiform scale and characteristic patches of superficially cracked skin (see Ch. 13).

## Traumatic Irritant Contact Dermatitis

*Traumatic* ICD may develop after acute skin trauma, such as from burns, lacerations, or acute ICD. Patients should be asked whether they have cleansed the skin with strong soaps or detergents. It is characterized by eczematous lesions, most commonly on the hands, that last for weeks to months with persistent redness, infiltration, scale, and fissuring in the affected areas.

## Pustular and Acneiform Irritant Contact Dermatitis

*Pustular and acneiform* ICD results from exposure to certain irritants, such as metals, croton oil, mineral oils, tars, greases, cutting and metal working fluids, and naphthalenes (see Table 15.3 and Ch. 16). This syndrome should be considered in conditions in which folliculitis or acneiform lesions develop in settings outside of typical acne, particularly in patients with atopic dermatitis, seborrheic dermatitis, or prior acne vulgaris. The pustules are "sterile" and transient. Miliarial reactions, which may become pustular, can develop in response to occlusive clothing, adhesive tape, or ultraviolet and infrared radiation.

## Non-erythematous Irritant Contact Dermatitis

*Non-erythematous* ICD may be defined as a subclinical form of ICD with early stages of skin irritation seen as changes in the stratum corneum barrier function without a clinical correlate.

## Subjective or Sensory Irritant Contact Dermatitis

*Subjective or sensory* ICD is characterized by reports of a stinging or burning in the absence of visible cutaneous signs of irritation. Irritants capable of eliciting this reaction include propylene glycol, hydroxy acids, ethanol, and topical medications such as lactic acid, azelaic acid, benzoic acid, benzoyl peroxide, mequinol, and tretinoin. Sorbic acid, a preservative in concentrations of up to 0.2% in foods, cosmetics, and drugs, may also produce sensory irritation in predisposed individuals. This reaction to irritants such as lactic or sorbic acid may be reliably reproduced with dose responsiveness in double-blinded exposure tests.

## Airborne Irritant Contact Dermatitis

*Airborne* ICD develops in irritant-exposed sensitive skin of the face and periorbital regions. While this often simulates photoallergic reactions, involvement of the upper eyelids, philtrum, and submental regions in patients with airborne ICD may aid in distinguishing between these two entities. Airborne ICD results from exposure to floating dusts, fibers (particularly fiberglass), fragrances, woods, and volatile solvents and sprays[32].

## Frictional Irritant Contact Dermatitis

*Frictional* ICD is a distinct ICD subtype resulting from repeated low-grade frictional trauma. It is often acknowledged to also play an adjuvant role in allergic contact dermatitis and ICD. The frictional response includes hyperkeratosis, acanthosis, and lichenification, often progressing to hardening, thickening, and increased toughness. It often involves the fingers of the dominant hand. Frictional ICD has been associated with repetitive handling of paper, carbon, fabric and plastic bags; driving (due to repetitive gripping of the steering wheel and gear stick); and rubbing from clothing or sports equipment, as in golf and fishing[33].

## Contact Urticaria

Contact urticaria is divided into non-immunologic and immunologic subtypes (see Ch. 16), with the former occurring more frequently and in the absence of previous exposure. Irritants that can produce *immunologic* contact urticaria include parabens (preservatives), henna, ammonium persulfate (oxidizing agent), and latex. *Non-immunologic* contact urticaria can be caused by a number of exposures, from caterpillars and jellyfish to stinging nettles and foods (see Table 15.3)[34]. Risk factors include atopy, hand dermatitis, previous mucosal exposure to latex (e.g. urinary catheterization), and allergies to fruits (e.g. kiwi, avocado; see Table 16.6).

# PATHOLOGY

The histopathologic features of ICD vary according to the nature of the irritant, but often include variable spongiosis, necrosis of keratinocytes, and an inflammatory infiltrate rich in neutrophils. The combination of a neutrophil-rich superficial perivascular infiltrate plus widely scattered necrotic keratinocytes is most typical; sometimes there is even full-thickness epidermal necrosis and subepidermal clefting. Mild ICD can resemble allergic contact dermatitis. Over time, additional histologic findings include acanthosis with mild hypergranulosis and hyperkeratosis. In aggregate, these elements are not specific, and they cannot be confidently differentiated from either chronic allergic contact dermatitis or other types of chronic eczemas[35].

# CLASSIFICATION OF IRRITANT CHEMICALS

## Acids

A variety of both inorganic and organic acids can be corrosive to the skin. Acids cause epidermal damage via protein denaturation and cytotoxicity. Principally, all strong acids give the same clinical features, including erythema, vesication, and necrosis (Fig. 15.2).

Inorganic acids are commonly used in industry, especially hydrofluoric, sulfuric, hydrochloric, chromic, nitric, boric, and phosphoric acids (Table 15.4). Hydrofluoric acid and sulfuric acid cause the most severe



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**Fig. 15.2 Chemical burn due to hydrochloric acid.** Development of erythema, edema, and bullae, some of which are hemorrhagic. *Courtesy Eugene Mirrer, MD.*

burns, even at low concentrations, and significant absorption may lead to systemic toxicity[36].

In general, organic acids tend to be less irritating. Among the organic acids, acetic, acrylic, formic, glycolic, benzoic, and salicylic acids are the most common irritants, particularly after prolonged exposure. The uses and properties of *acrylic acid* and *formic acid* are outlined in Table 15.4. *Acetic acid* is a constituent of vinegar, flavoring agents, and astringent mouthwashes, whereas *glycolic*, *benzoic*, and *salicylic acids* are mild irritants whose properties can be harnessed for therapeutic and cosmetic purposes, when used in low concentrations.

## Alkalis

Alkalis or bases often cause more painful and severe damage than most acids, with the exception of hydrofluoric acid. There are generally no vesicles, but rather necrotic skin that first appears dark brown, then black, and ultimately becomes hard, dry, and cracked. Alkalis disrupt barrier lipids and denature proteins with subsequent fatty acid saponification, thus subjecting the cell to edema and resultant cytotoxicity. The emulsifying effect of soaps formed in the process facilitates the further penetration of the alkali into the deeper layers of the skin. Strong alkalis include sodium, ammonium, calcium, and potassium hydroxide; sodium and potassium carbonate; and calcium oxide, used primarily in the manufacture of bleaches, dyes, vitamins, pulp, paper, plastics, and soaps and detergents[37,38]. Calcium hydroxide is liberated from wet cement, which has an initial pH of 10–12 that rises to 12–14 as the cement sets (see Table 15.4).

## Metal Salts

Metals represent an important and commonly encountered group of irritants, with irritant reactions to metal salts ranging from folliculitis and pigmentary changes to ulceration (see Table 15.4).

## Solvents

A wide variety of solvents are used daily in processes such as chemical reactions, hydraulic systems, metal refining, dry cleaning, and metal degreasing. Nearly all are primary irritants to varying degrees (see Table 15.4), with only a few, such as turpentine, also being able to elicit allergic sensitization. Solvents act mainly by dissolving the intercellular lipid barrier of the epidermis, thereby substantially compromising barrier function. Prolonged skin contact can result in severe dermatitis as well as symptoms and even death from systemic absorption, making early recognition of skin manifestations important in the prevention of systemic toxicity. After repeated exposure, the hands, and occasionally the hands and face, develop erythema, scaling and dryness, eventually evolving into eczema (Fig. 15.3). The irritating capacity of organic solvents, attributed mainly to their lipophilicity, follows the order: aromatic > aliphatic > chlorinated > turpentine > alcohols > esters > ketones[39]. In the common clinical scenario of solvent then soap and water exposure, there is a synergistic degradation of epidermal barrier function, pointing to the importance of personal protection and hygiene[40].

## Alcohols/Glycols

*Alcohols* are used widely as solvents, disinfectants, preservatives in cosmetics, and penetration enhancers in drug delivery systems; most have only mild irritating effects. Alcohols are the safest known topical antiseptic compounds, providing bactericidal activity against most Gram-positive and Gram-negative bacteria as well as many fungi and viruses. Most appropriate for this use are diluted solutions of ethyl alcohol, propyl alcohol, and isopropyl alcohol, which act by means of protein denaturation. In cosmetics, alcohol is used as a preservative to prevent microbial contamination and to decrease viscosity. The principal mechanism by which alcohols enhance percutaneous absorption is hypothesized to be the extraction of intercellular lipids from the stratum corneum[41,42].

*Glycols*, or diols, such as ethylene glycol and propylene glycol, are aliphatic alcohols commonly used in cosmetic products as solvents, emulsifiers, humectants, or keratolytics. Propylene glycol can produce both allergic and irritant contact dermatitis and sources of exposure include personal care products, topical corticosteroids, and other topical medications[43]. Of note, propylene glycol is typically used in cosmetics in concentrations <50%[44].

## Detergents and Cleansers

A detergent includes almost any surface-active agent (surfactant) that concentrates at oil–water interfaces and holds both cleansing and emulsifying properties. Detergents most commonly cause chronic forms of ICD and are present in skin cleansers, cosmetics, and household cleaning products. With normal use, ICD is rare from skin cleansers, the exception being in individuals with susceptible skin. Detergents' cleansing actions are derived from their ability to lower the surface tension between two non-mixable phases due to their hydrophilic (polar head) and lipophilic (apolar tail) components. Skin toxicity arises from their damaging influence on the stratum corneum, which impairs barrier function (Fig. 15.4). Surfactants concomitantly bind to keratin and cause protein denaturation. When the stratum corneum is disrupted, detergents can damage viable epidermis and papillary dermal structures.

Irritancy from detergents is best evaluated by measuring TEWL since the latter represents the surrogate measurable change indicative of irritancy damage from this group of chemicals. As a group, anionic detergents such as alkyl sulfates and alkyl carboxylate salts (soap) are more irritating than are amphoteric and non-ionic groups. Sodium lauryl sulfate, an anionic detergent, is often used as a reference irritant in studies because of its consistent, non-allergic, rapid toxic response. Amphoteric surfactants such as *cocamidopropyl betaine* – which are used in therapeutic formulations, personal care products, and cosmetics – have the lowest irritation potential[45,46]. *Cocamide DEA* (diethanolamine), a non-ionic biodegradable surfactant, is used as a viscosity booster, stabilizer, and foam booster; it is found in hand soaps, liquid shampoos, detergents, and dishwashing liquids. Of note, cocamide DEA is one of the more irritating surfactants[47], and it is commonly involved in occupation-related contact dermatitis in North American healthcare workers[48]. In contrast, *cocamide MEA* is minimally irritating and can be used in a concentration of up to 10% in leave-on products[49].

Skin cleansers may be solid or liquid, based on soap and/or synthetic detergents, and may contain solvents or abrasives, depending on use requirements (see Ch. 153). Although the primary factor that determines skin irritancy is the detergent component, skin tolerance cannot be adequately predicted from the composition of products alone. Such an assessment is largely left to trial and error or repeated skin challenges in volunteer cohorts.

## Disinfectants

Most disinfectants used to destroy pathogens in the environment act as weak toxic agents and cause chronic ICD as a result of cumulative doses of subclinical irritancy (Fig. 15.5). Various compounds may be used, such as alcohols (see above), aldehydes, phenolic compounds, halogenated



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| IRRITANT CHEMICALS: USES, PROPERTIES AND SIDE EFFECTS | | |
|---|---|---|
| **Compound** | **Uses/exposures** | **Properties and side effects** |
| **Inorganic acids** | | |
| Chromic acid | • Metal treatments (chrome plating, copper stripping, aluminum anodizing) | • Ulcerations known as "chrome holes"<br>• Nasal septal perforation |
| Hydrochloric acid | • Production of fertilizers, dyes, paints, and soaps<br>• Electroplating, oil refining, and food processing | • Erythema, vesication, and necrosis (Fig. 15.2) |
| Hydrofluoric acid | • Semiconductor industry (etching glass, metal, and stone; can dissolve silicon)<br>• Household and commercial rust, stain, and lime-scale removers | • Extremely cytotoxic agent that can lead to severe tissue damage<br>• Penetrates intact skin due to its limited dissociation constant<br>• Painful penetration may continue for days after exposure<br>• With low concentrations, the onset of symptoms may be delayed for up to a day<br>• Intense pain is due to the capacity of fluoride ions to bind tissue calcium, thereby affecting nerve conduction<br>• Hypomagnesemia and a fluoride-mediated increase in pulmonary vascular resistance may also play a role in toxicity |
| Nitric acid | • Production of fertilizers and explosives; metal etching and cleaning<br>• Fuming nitric acid – a chemical intermediate used in rocket fuels and as a laboratory reagent | • Chemical burns<br>• Irritant contact dermatitis<br>• Distinctive yellow discoloration<br>• Fuming nitric acid – highly corrosive |
| Phosphoric acid | • Phosphate fertilizer, pharmaceuticals, water treatment, animal feeds, soaps and detergents, and cotton dyeing | • Erythema, vesication, and necrosis |
| Sulfuric acid | • Manufacture of fertilizers, inorganic pigments, textile fibers, explosives, pulp, and paper<br>• Occupational risk: jewelers, electroplaters, metal cleaners, and storage battery makers | • Erythema, vesication, and necrosis |
| **Organic acids** | | |
| Acrylic acid | • Monomer for acrylic plastics, synthetic resins, dentures, artificial nails, adhesives, and paints | • Highly irritating and corrosive<br>• Capable of penetrating rubber gloves<br>• Also allergic contact dermatitis |
| Formic acid | • Neutralizer in leather manufacturing, an antiseptic in brewing, and a coagulant in the production of natural latex | • Greatest corrosive potential of the organic acids |
| **Alkalis** | | |
| Cement (mixed with water) | • Construction | • Acute ulcerative damage as a result of the high alkalinity of wet cement (due to liberation of calcium hydroxide)<br>• Necrotic skin changes appear 8–12 hours after exposure<br>• Chronic irritant cement dermatitis – develops over months to years from continued exposure; can accompany allergic contact dermatitis to chromium<br>• Most common locations – lower limbs (kneeling, contaminated footwear), hands (glove contamination) |
| **Metal salts** | | |
| Arsenic compounds (including arsenic trioxide) | • Aerosolized during the smelting of copper, gold, lead and other metals<br>• Manufacture of pesticides and herbicides<br>• Semiconductor industry and smelting operations | • Persistent folliculitis<br>• Potential for systemic toxicity (see Ch. 88) |
| Beryllium | • Production of hard, corrosion-resistant alloys | • Ulcerated granulomatous skin lesions (local reaction; see Ch. 94)<br>• Sarcoidosis-like granulomas of the skin and lung (systemic reaction) |
| Calcium oxide (quicklime) | • Manufacture of pulp, steel, and paper | • Releases heat on contact with water, causing painful cutaneous ulcerations |
| Cobalt salts | • Alloys, ceramics, electroplating, electronics, magnets, paints, varnishes, and hair dyes | • Irritant and allergic contact dermatitis |
| Copper salts (e.g. sulfates, oxides, cyanides) | • Used as a biostatic surface and in wood preservatives and fungicides | • Impart a greenish-black color to hair, skin, and teeth if persistent exposure to dust forms<br>• Inhalation of copper oxides (as well as magnesium and zinc oxides) can cause metal fume fever, a brief influenza-like illness usually associated with welding operations |

**Table 15.4 Irritant chemicals: uses, properties and side effects.**



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| IRRITANT CHEMICALS: USES, PROPERTIES AND SIDE EFFECTS | | |
|---|---|---|
| **Compound** | **Uses/exposures** | **Properties and side effects** |
| Mercury, inorganic | • Products containing inorganic mercury have been banned in US since 1990<br>• Imported skin-lightening creams*<br>• Limited exposure possible from old cans of latex paint | • Bluish linear pigmentation of the tongue and gums, which may serve as a marker for systemic mercury poisoning |
| Mercury, organic | • Phenyl mercury salts are used as preservatives in cosmetics and vaccines, as fungicides, herbicides and pesticides in agriculture, and in dental restorative materials | • Although once used as skin disinfectants, both irritant and allergic contact dermatitis have restricted its use<br>• Positive patch tests to mercury (as well as other metals such as gold) observed in patients with oral lichenoid reactions to dental materials (see Ch. 11) |
| Selenium compounds (including sulfide) | • Therapeutic shampoo (selenium sulfide) | • Skin irritation, conjunctivitis |
| Thimerosal | • Topical medicaments, vaccines, and cosmetics | • Common cause of both allergic and irritant patch-test reactions |
| **Solvents** | | |
| Benzene | • Manufacture of polymers, plastics, resins, adhesives, rubber, lubricants, dyes, detergents, drugs, explosives, and pesticides | • Petechial eruption of the trunk, extremities, and mucous membranes<br>• Petechiae considered to be a marker for aplastic anemia |
| Chlorinated hydrocarbons (e.g. carbon tetrachloride, trichloroethylene, tetrachloroethane, methylene chloride, ethylene chloride) | • Production of vinyl chloride, which is then converted to polyvinyl chloride<br>• Pesticides<br>• Dry cleaning | • Cutaneous irritation<br>• Systemic effects: liver and kidney damage, CNS depression, and carcinogenesis<br>• Methylene chloride can hydrolyze to form hydrochloric acid, thus leading to significant cutaneous irritation (see above)<br>• "Degreaser's flush" – sudden erythema in extensive areas of the face, neck, and shoulders; caused by alcohol ingestion shortly before or during inhalation of trichloroethylene |
| Coal tar derivatives (e.g. toluene, xylene, ethyl benzene, cumene) | • Medications<br>• Solvents | • Skin drying and defatting<br>• Exposure solely to vapors can cause xerosis |
| Petrochemicals (e.g. gasoline, hexane, kerosene, Stoddard solvent) | • Fuels<br>• Insecticides<br>• Dry cleaning<br>• Hexanes – gasoline, glues, and for extraction of cooking oils from seeds | • Irritant contact dermatitis<br>• Absorption of hexane via inhalation or percutaneously can cause paresthesias, hypoesthesia, and motor weakness |
| **Disinfectants** | | |
| Aldehydes (e.g. formaldehyde, glutaraldehyde, hexamethylene-tetramine) | • Disinfectants, biocides<br>• Tissue fixative, embalming agent<br>• Adhesives, resins | • More irritating than alcohols<br>• Formaldehyde – high chemical reactivity with proteins; both allergen and irritant, even at low concentrations |
| Ethylene oxide | • Sterilization of medical equipment | • Oxidizing agent<br>• Severe chemical burns |
| Halogens – Chlorines | • Disinfectants, including wounds†<br>• Bleach (sodium hypochlorite)<br>• Sanitizers for swimming pools, hot tubs | • Denature proteins via deaminating or chlorinating amino acids<br>• Oxidizing agent (sodium hypochlorite)<br>• Acute skin and mucosal irritant reactions, particularly at high temperatures |
| Halogens – Iodines | • Broad-spectrum antimicrobial and sporicidal agents<br>• Surgical scrub, shampoo, and skin cleanser (povidone-iodine in a surfactant base) | • Inhibit DNA, RNA, and protein synthesis<br>• Irritant contact dermatitis<br>• Contact urticaria |
| Phenolic compounds (e.g. Lysol® [cresol and soap solution], pentachlorophenol, chloroxylenol) | • Disinfectants<br>• Preservative in over-the-counter products (baby powders, shampoos), especially chloroxylenol | • Irritant contact dermatitis<br>• Chemical leukoderma (amyl- and butyl-phenol compounds; see Table 15.3) |
| Quaternary ammonium salts (e.g. benzalkonium chloride) | • Cosmetics, medications (e.g. ophthalmic solutions)<br>• Antiseptics (e.g. cleaning of surgical instruments)<br>• Detergent | • Cationic surfactants that can precipitate or denature proteins and destroy microorganisms<br>• Local skin irritation depends on the solution concentration<br>• Benzalkonium chloride – irritant reactions frequent during patch testing (concentration, 0.1%); can also be an allergen (e.g. healthcare workers, leg ulcer patients) so clinical correlation required |

*Rarely organic mercury.
†Dakin's solution contains sodium hypochlorite; EUSOL (Edinburgh solution of lime) contains chlorinated lime and boric acid.

**Table 15.4 Irritant chemicals: uses, properties and side effects.** Glycols/alcohols and detergents/cleansers are discussed in the text.



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**Fig. 15.3 Bilateral irritant contact dermatitis of the palms secondary to repeated contact with paint solvents.** Extensive patch testing excluded allergic contact dermatitis in this professional paint and crayon illustrator. *Courtesy Kalman Watsky, MD.*

**Fig. 15.4 Irritant contact dermatitis caused by playing with slime composed of school glue, food coloring, and laundry detergent.** Note the involvement of the web spaces. Components of homemade slime typically include glue, dye, and sodium borate (borax powder or within detergents) or boric acid. *Courtesy Emily Berger, MD.*

compounds, and quaternary ammonium salts (see Table 15.4), in addition to dyes, oxidizing agents, and mercury compounds[50–52].

*Dyes* of the triphenylmethane family are extensively used as topical antiseptics; they are capable of producing phototoxic dermatitis but uncommonly cause irritant reactions. Benzoyl peroxide is a common *oxidizing agent* used as a topical antimicrobial medication that is capable of causing mild irritation. Most *mercury* compounds induce skin irritation by protein precipitation, but have fallen into disuse as a result of hazards of systemic toxicity.

## Plastics

Plastics are synthetic macromolecular end-products consisting of large polymers formed by the linkage of small monomers into large chain-like units. A large number of plastic resins are commercially important and they can be divided into three categories:

- thermoplastics – polyacrylates, polyethylene, polystyrene, polyvinyl chloride, and saturated polyesters
- thermosettings – epoxy resins, phenol–formaldehyde resins, and polyurethanes
- elastomers – synthetic rubbers.

Skin damage is almost exclusively attributable to the monomer ingredients, hardeners, and other additives such as stabilizers. The final hardened plastic product is generally considered inert and non-hazardous to

**Fig. 15.5 Irritant contact dermatitis of the hands due to chronic exposure to disinfectants.** *Courtesy Antonio Torrelo, MD.*

the skin, but residual non-polymerized monomers may be the offending agents. Both irritant and allergic contact dermatitis are common, and they may only be differentiated by patch testing[53].

## Food and Plants

Food and food additives often contain compounds that elicit contact sensitivity and irritation, particularly in those working in agriculture, fishing, catering, and food processing industries. Much of the work in these sectors is done without gloves under damp working conditions with frequent hand washing – factors further aggravating skin irritation. Mechanical, thermal, and climatic factors also contribute. A large majority of exposed persons in food handling and fishing professions may be affected by chronic irritant hand dermatitis[54]. Slight irritant skin changes are often accepted as "normal" for the patient's occupation and medical advice is often not sought.

Examples of foods that can lead to ICD include pineapples, which contain the proteolytic enzyme bromelain (bromelin), and garlic, which contains allicin and diallyl disulfide. Irritant dermatitis and chemical burns have occurred from the homeopathic use of garlic as well as apple cider vinegar (contains acetic acid; see Fig. 90.7)[55]. Foods can also lead to contact urticaria (see Table 15.3) and protein contact dermatitis; both of these entities are discussed in detail in Chapter 16.

Additional food-associated and plant-derived irritants are outlined in Table 15.5 and the latter are discussed in Chapter 17.

## Water

Water, the universal solvent, is a particularly ubiquitous skin irritant. Wet workers such as hairdressers, hospital cleaners, cannery workers, and bartenders have an increased incidence of hand eczema, with frequent exposure to water via hand washing directly correlating with the development of irritant hand dermatitis[24]. Several mechanisms such as osmolarity, pH, hardness, and temperature might account for the irritancy of water. Stratum corneum hydration may facilitate penetration of polar and non-polar substances through connections in the lacunar network (see Ch. 124) as well as support the overgrowth of pathogenic organisms. The irritancy of water is often exacerbated by occlusion, as it changes the barrier properties of the stratum corneum[56].

## Bodily Fluids

Urine, feces (especially in the setting of diarrhea), and saliva can lead to ICD. In babies, irritant diaper dermatitis is a common problem and is often characterized by glazed erythema of convex surfaces and at the diaper margins, with sparing of the skin folds; edema, scaling, and superficial erosions may also be observed (see Fig. 13.12). Chronic diarrhea and incontinence can result in similar problems in the elderly[57].



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| CHEMICAL COMPOUNDS IN PLANTS RESPONSIBLE FOR CHEMICAL IRRITANT CONTACT DERMATITIS | | | |
|---|---|---|---|
| **Irritant** | **Mechanism of action** | **Clinical features** | **Example of plants** |
| Bromelain | Proteolytic activity | Cutaneous irritation | Pineapple |
| Capsaicin | Binds to vanilloid receptors on sensory neurons | Cutaneous burning and erythema | Hot pepper |
| Isothiocyanates (mustard oils) | Readily react with amino groups of proteins in the skin and mucosae | Short-lived cutaneous burning and erythema; urticarial<br>Ocular irritation | Mustard, cabbage, brussels sprouts, broccoli, cauliflower, radishes |
| Nicotine | Antagonist for nicotinic receptors in the peripheral and central nervous systems | Cutaneous irritation | Tobacco |
| Phorbol esters | Bind to phospholipid-dependent membrane receptors | Cutaneous pain, erythema, edema, and blistering<br>Conjunctivitis and even blindness | Texas croton, castor bean |
| Protoanemonin (ranunculol) | Disrupts disulfide bonds, resulting in subepidermal separation | Cutaneous burning and erythema<br>Irritation of conjunctivae and nasal airway | Buttercups, columbine (when wounded) |

**Table 15.5 Chemical compounds in plants responsible for chemical irritant contact dermatitis.** For additional causes, see Table 17.6.

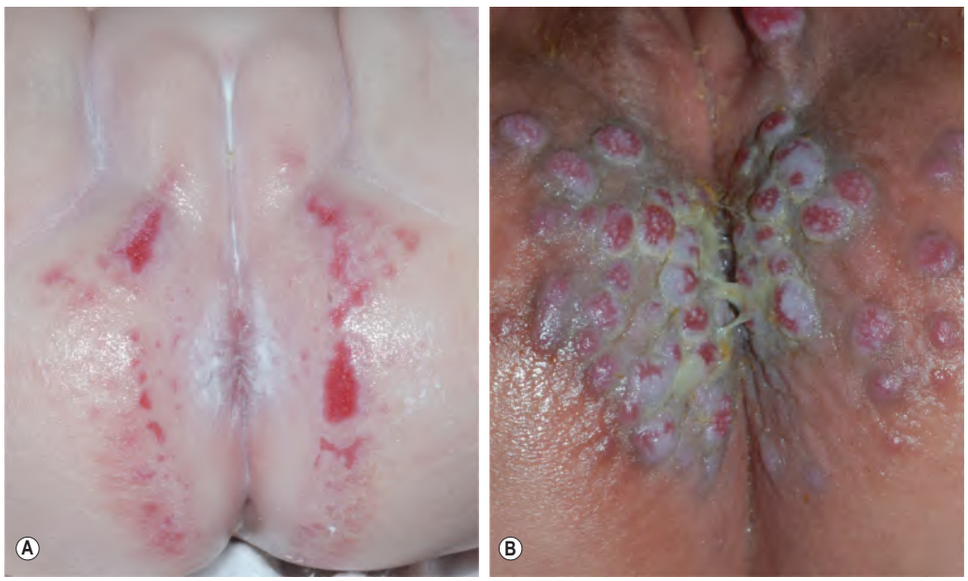


**Fig. 15.6 Granuloma gluteale infantum, a form of chronic irritant contact dermatitis.** Chronic exposure to urine and feces can lead to erosions and ulcerations (**A**); also referred to as Jacquet erosive dermatitis, as well as pseudoverrucous papules and nodules or erosive papulonodular dermatosis (**B**). *Courtesy Julie V. Schaffer, MD.*

In both children and adults, long-standing irritant dermatitis in the anogenital region can lead to papules and nodules (granuloma gluteale infantum [adultorum]; pseudoverrucous papules and nodules) as well as punched-out erosions and ulcerations (Jacquet erosive dermatitis) (Fig. 15.6)[58]. Predisposing factors, in addition to prolonged exposure to urine or stool, include occlusion, candidal infection, and topical corticosteroid use. Histologically, epidermal hyperplasia, a variably dense mixed inflammatory infiltrate in the dermis, and vascular proliferation are typically observed. The term erosive papulonodular dermatosis has been suggested as an overarching term to describe this spectrum of reactive changes. Of note, similar changes can be seen in submammary and peristomal skin. Although barrier creams and topical antifungal agents may have some benefit, the condition tends to persist until the irritant trigger is eliminated.

Children, more so than adults, develop the habit of lip licking. In these patients, irritant dermatitis may involve the perioral skin as well as the lips (Fig. 72.16A). In a series of 75 patients with cheilitis who were referred for patch testing, ICD was the most common cause of cheilitis[59,60]. Figure 72.15 outlines distinguishing features of the various forms of cheilitis, including that due to ICD. Periurethral ICD may be seen in patients with bladder-drained pancreatic allografts and in those receiving foscarnet.

## DIFFERENTIAL DIAGNOSIS

Despite their different pathogeneses, allergic and irritant contact dermatitis, especially of the chronic type, show a remarkable similarity with respect to clinical appearance, histopathology, and immunohistology. Clinically, the reactions often look identical, with erythematous plaques, xerosis, scaling, fissures, and lichenification locally distributed and sharp borders delineating the areas of contact. In the clinical setting, the substance in question is often unknown, as is the concentration and duration of exposure. As a result, the diagnosis of ICD has remained a diagnosis of exclusion – when the dermatitis cannot be explained by a positive patch test to a known allergen. Another helpful, but not conclusive, clinical feature is the more frequent complaint of burning, pain, and stinging with ICD, in contrast to pruritus in areas of allergic contact dermatitis (Fig. 15.7).

## PROGNOSIS

In many individuals, ICD resolves spontaneously even with continuous exposure, a process referred to as "accommodation" or "hardening"[61]. The exact mechanisms are still unclear, but the following changes have been observed after repetitive cutaneous exposure to irritants:



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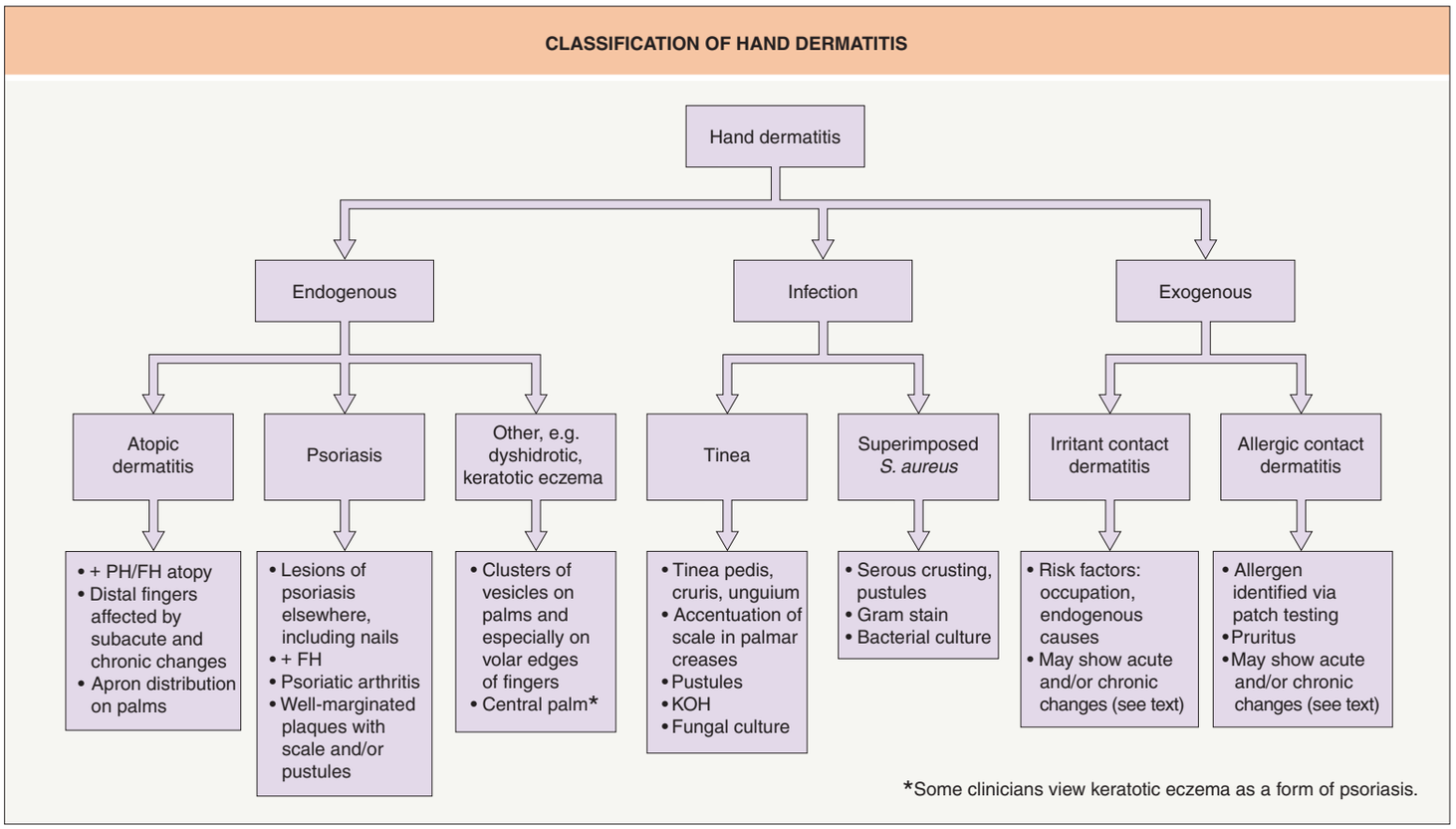


**Fig. 15.7 Classification of hand dermatitis.** More than one etiology may be present, e.g. atopic dermatitis plus irritant contact dermatitis. Keratolysis exfoliativa is usually not confused with hand dermatitis. Also see Figure 17.1. FH, family history; PH, personal history. (For further details, see Seidenari S, Giusti F, Pepe P, Mantovani L. Contact sensitization in 1094 children undergoing patch testing over a 7-year period. *Pediatr Dermatol.* 2005;22:1–5.)

- improvement of the physical barrier via formation of a thicker stratum corneum and a thicker stratum granulosum as well as increased production of ceramide 1
- increased skin permeability to irritants plus changes in vascular reactivity that allow faster removal of irritants
- immunologic alterations that favor an anti-inflammatory response to irritants, e.g. increased ratio of IL-1Ra (an anti-inflammatory cytokine) to IL-1α (a proinflammatory cytokine)
- a systemic hyporeactive state following repetitive exposure to low-dose irritants[61].

The time required for "accommodating" to the aggravating stimulus varies among individuals, and for some individuals, ICD can be a chronic and devastating problem. Poor prognosis is related to a previous history of atopy, female sex, and the presence of allergic contact dermatitis (as well as ICD). Factors that can potentially improve prognosis are early diagnosis, treatment, and patient knowledge about the disease[62].

## TREATMENT

Avoidance of causative irritants in the home or the workplace is the primary treatment for ICD. Strategies in the prevention of ICD include the identification of irritants with appropriate substitution, the establishment of engineering controls to reduce exposure, the utilization of personal protective equipment such as gloves and special clothing, and barriers such as ointments, emollients, or creams. Other preventive strategies include emphasizing personal and occupational hygiene, establishing educational programs to increase awareness in the workplace, and providing health monitoring.

In the workplace, by far the most effective measure for reducing the incidence of contact dermatitis is technical avoidance. This can be accomplished via shielding and personal protection of the workers and limiting the use of potent irritants to closed or automated systems. Preventive skin care at the workplace may mitigate occupational hand dermatitis among vulnerable populations and involves regular use of protective creams to intact skin, removal of irritants by mild cleaning agents, and enhancement of barrier function generation by emollients or moisturizers[63,64]. Non-irritating fatty substances such as petrolatum preclude hydrophilic chemical penetration and restore barrier function. The efficacy of barrier creams for skin protection (as compared to bland emollients) remains a topic of debate[65]. In a randomized controlled trial, both barrier creams and moisturizing vehicles were found to positively influence skin status and hydration, without significant differences in efficacy[66]. Newer moisturizers containing ceramides can also improve barrier function, but further trials are necessary to ascertain their efficacy compared to traditional moisturizers[67,68]. Lastly, while new bioengineering techniques have been proposed for the assessment of individual irritant sensitivity, their value in predicting occupational ICD, especially of the cumulative type, remains unclear.

Protective gloves can reduce risk of irritant hand dermatitis when appropriately selected and worn; however, gloves worn incorrectly or produced with poor quality control can actually instigate or worsen ICD. Glove failure can result in penetration, permeation, or contamination of hazardous materials, with irritation further compounded by the gloves' occlusive environment[69,70].

The establishment of appropriate educational prevention programs is essential. A project in Finland showed a significantly better outcome for employees with occupational hand dermatitis attending an "eczema school"-like clinic run by a specialized nurse compared with an unschooled control group[71]. Another study found that most occupational skin diseases responded to effective secondary preventive measures, combining employee medical treatment with exposure analysis-based individual and group training in preventive measures[72]. Education was important in making the employees aware of initial skin changes, such as slight erythema and scaling in the interdigital folds, indicating the need to optimize skin protection and care measures in order to prevent exacerbation and chronicity.

The goal of treatment is to restore normal epidermal barrier function. Topical corticosteroids are frequently used, but their efficacy has



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been controversial, as experimental studies have provided conflicting results[73]. In one double-blind, vehicle-controlled study, statistically lower values of erythema and TEWL were observed in sites irritated by sodium lauryl sulfate after 7 days of treatment with betamethasone valerate[74]. Systemic corticosteroids, although potentially helpful in reducing acute inflammation, are not useful in the treatment of chronic ICD unless corrective measures are taken to avoid the offending contactants. Narrowband ultraviolet B or photochemotherapy (PUVA) irradiation may be considered for chronic dermatitis that does not respond to any other form of therapy. Hyperkeratotic palmoplantar dermatitis from frictional or chronic ICD or a combination of dermatitis and psoriasis may benefit from the adjunctive use of systemic retinoids such as acitretin and alitretinoin or systemic immunomodulators such as methotrexate, cyclosporine, and possibly targeted immunomodulatory (biologic) therapy[75].

**Additional figures and tables on Causes of cheilitis in patients who were referred for patch testing, The Hand Eczema Severity Index (HECSI), and The modified Total Lesion Symptom Score (mTLSS) for hand eczema, available in our eBook (see inside front cover for access code).**

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**eFig. 15.1 Moderately severe irritant contact dermatitis of the hands due to chronic exposure to disinfecting solutions and antiseptics.** The results of patch testing, latex challenge testing, and RAST testing were negative in this practicing dentist.

**eFig. 15.3 Irritant contact diaper dermatitis (Jacquet erosive dermatitis).** Multiple punched out erosions and ulcerations. *Courtesy Julie V. Schaffer, MD.*

**eFig. 15.2 Chemical burn from topical application of apple cider vinegar.** There is sparing of the skin folds and this may be misdiagnosed as child abuse (see Ch. 90). *Courtesy Julie V. Schaffer, MD.*

**eFig. 15.4 Irritant contact diaper dermatitis (granuloma gluteale infantum).** Coalescing moist, pink papules on the vulva and suprapubic area of an infant. *Courtesy Julie V. Schaffer, MD.*



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| CAUSES OF CHEILITIS IN PATIENTS REFERRED FOR PATCH TESTING | |
|---|---|
| **Irritant contact dermatitis**<br>Most frequent cause: lip licking | 36% |
| **Allergic contact dermatitis**<br>Most common allergens (% of patients in a second series[60]): fragrance mix, 30%; *Myroxylon pereirae* (balsam of Peru), 23%; nickel, 22%; gold, 13%; neomycin, 12%; cobalt, 10%; propylene glycol, 8%; lanolin, 7%; cinnamic aldehyde, 7%; bacitracin, 5%; benzophenone, 5%; methyldibromo glutaronitrile/phenoxyethanol, 5%; tea tree oil, 5%; budesonide, 5% | 25% |
| **Atopic dermatitis** | 19% |
| **Eczema** (cause unknown) | 9% |
| **Non-eczematous causes** (angioedema, photosensitivity, psoriasis, erosive candidal cheilitis, functional) | 9% |
| **Seborrheic dermatitis** | 1% |

**eTable 15.1 Causes of cheilitis in patients who were referred for patch testing (n = 75).** *Adapted from Freeman S, Stephens R. Cheilitis: analysis of 75 cases referred to a contact dermatitis clinic. Am J Contact Dermat. 1999;10:198–200 and Zug KA, Kornik R, Belsito DV, et al.; North American Contact Dermatitis Group. Patch-testing North American lip dermatitis patients: data from the North American Contact Dermatitis Group, 2001 to 2004. Dermatitis. 2008;19:202–8.*

| THE HAND ECZEMA SEVERITY INDEX | | | | | |
|---|---|---|---|---|---|
| **Clinical signs** | **Location and intensity score** | | | | |
| | *Fingertips* | *Fingers (except tips)* | *Palm of hands* | *Back of hands* | *Wrists* |
| Erythema | 0–3 | 0–3 | 0–3 | 0–3 | 0–3 |
| Infiltration/ papulation | 0–3 | 0–3 | 0–3 | 0–3 | 0–3 |
| Vesicles | 0–3 | 0–3 | 0–3 | 0–3 | 0–3 |
| Fissures | 0–3 | 0–3 | 0–3 | 0–3 | 0–3 |
| Scaling | 0–3 | 0–3 | 0–3 | 0–3 | 0–3 |
| Edema | 0–3 | 0–3 | 0–3 | 0–3 | 0–3 |
| **Intensity sum** (range) | 0–18 | 0–18 | 0–18 | 0–18 | 0–18 |
| **Extent score** of clinical symptoms (Ex) | 0–4 | 0–4 | 0–4 | 0–4 | 0–4 |
| Total HESCI Score = | Intensity sum × Ex + | Intensity sum × Ex + | Intensity sum × Ex + | Intensity sum × Ex + | Intensity sum × Ex |

**eTable 15.2 The Hand Eczema Severity Index (HECSI).** The HECSI evaluates hand eczema severity based on 6 clinical signs and 5 locations. Each location is given a score from 0 to 3 for the *intensity* of clinical symptoms, graded on the following scale: *0, no skin changes; 1, mild disease; 2, moderate; and 3, severe*. For each location (in consideration of both hands) the affected area is given a score from 0 to 4 for *extent* of clinical symptoms (*0 = 0%; 1 = 1%–25%; 2 = 26%–50%; 3 = 51%–75% and 4 = 76%–100%)*. The score given for the extent for each location is multiplied by the total sum of the intensity of each clinical feature. The total score ranges from 0 to a maximum severity score of 360. A composite HECSI score of 0 to 11 corresponds to mild disease; scores of 12 to 27 correspond to moderate disease; scores of 28 or more correspond to severe disease. *Adapted from Held E, Skoet R, Johansen JD, Agner T. The hand eczema severity index (HECSI): a scoring system for clinical assessment of hand eczema. A study of inter- and intraobserver reliability. Br J Dermatol. 2005;152(2):302-307; and Oosterhaven J a. F, Schuttelaar MLA. Responsiveness and interpretability of the hand eczema severity index. Br J Dermatol. 2020;182(4):932-939.*

| THE MODIFIED TOTAL LESION SYMPTOM SCORE (MTLSS) FOR HAND ECZEMA | |
|---|---|
| **Parameter** | **Severity score*** |
| Erythema | 0 = Absent<br>1 = Faint erythema<br>2 = Prominent redness<br>3 = Deep intense red color |
| Scaling | 0 = Absent<br>1 = Slight flaking over limited areas, mostly fine scales<br>2 = Flaking over widespread area(s), coarser scales<br>3 = Desquamation covering over 30% of the hand, with coarse thick scales |
| Lichenification/ hyperkeratosis | 0 = Absent<br>1 = Mild thickening with exaggerated skin lines over limited areas<br>2 = Palpable thickening over widespread area(s)<br>3 = Prominent thickening over widespread area(s) with exaggeration of normal skin markings |
| Vesiculation | 0 = Absent<br>1 = Scattered vesicles affecting up to 10% of hand, without erosion<br>2 = Scattered or clustered vesicles affecting up to 30% of hand, without visible erosion or excoriation |
| Edema | 0 = Edema<br>1 = Dermal swelling over less than 10% of hands<br>2 = Definite dermal swelling over more than 10% of hand<br>3 = Dermal swelling with skin induration over widespread area(s) |
| Fissures | 0 = Absent<br>1 = Cracked skin affected a small area of the hand<br>2 = Cracked skin affecting multiple areas of the hand and causing pain<br>3 = One or more deep fissures and causing bleeding or severe pain |
| Pruritus/pain | 0 = Absent<br>1 = Occasional, slight discomfort a few times per day<br>2 = Intermittent, causing discomfort frequently during the day<br>3 = Persistent or interfering with sleep |
| *1 = mild; 2 = moderate; 3 = severe. | |

**eTable 15.3 The modified Total Lesion Symptom Score (mTLSS) for hand eczema.** Composite scores range from 0 (no signs or symptoms) to 21 (more serious disease). *Adapted from Ruzicka T, Lynde CW, Jemec GBE, et al. Efficacy and safety of oral alitretinoin (9-cis retinoic acid) in patients with severe chronic hand eczema refractory to topical corticosteroids: results of a randomized, double-blind, placebo-controlled, multicentre trial. Br J Dermatol. 2008;158:808–17.*



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